

EXHIBIT 28
(Public Version)

Ex. 28C - Domestic Industry Claim Chart
U.S. Patent No. 12,361,760 (“the ’760 patent”)

trinamiX GmbH (“trinamiX”) hereby identifies evidence demonstrating the following trinamiX products practice the ’760 Patent: all trinamiX products that include trinamiX Face Authentication, including but not limited to e.g., the trinamiX Face Authentication software libraries and SDK, ET-90 demonstrator, Cockatiel demonstrator, and MPIC2.0 (collectively, the “Practicing ’760 Products”). These products are made by trinamiX, and are imported, sold for importation, and/or sold within the United States after importation by trinamiX. The chart below is based on evidence of representative products of the Practicing ’760 Products. Each of the following limitations is met literally, and, to the extent a limitation is not met literally, it is met under the doctrine of equivalents.

The trinamiX products identified in this chart are representative of all relevant trinamiX products containing relevant components practicing the patent. The same or similar analysis as discussed below apply to all relevant trinamiX products containing relevant components practicing the patent because, for example, the cameras, sensors, and software operate similarly within each device, and each corresponding device has the same or similar components, features, functions, and/or characteristics. This will further be confirmed through expert discovery.

Claim 1	Practicing ’760 Products
[1pre] A method for face authentication comprising the following steps:	To the extent the preamble is limiting, the Practicing ’760 Products comprise a method for face authentication.

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	A large solid black rectangular redaction box covering the entire right-hand cell of the top row.
[1a] at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined	The Practicing ’760 Products comprise at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit. The Practicing ’760 Products comprise a processing unit, such as a Qualcomm chip processor, and a camera, such as a CMOS camera, which identifies geometrical feature characteristic for faces in captured 2D images:

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or pre-determined geometrical
feature characteristic for faces by
using at least one processing unit;



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[1b1] at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit and determining at least one second image using the at least one camera,	The Practicing ’760 Products comprise at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit and determining at least one second image using the at least one camera. The Practicing ’760 Products comprise an illumination unit, such as a dot projector and flood illuminator, which projects illumination patterns for skin detection: 

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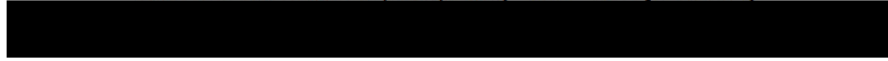
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	A large black rectangular area covering the majority of the table's content, indicating that the information has been redacted.	
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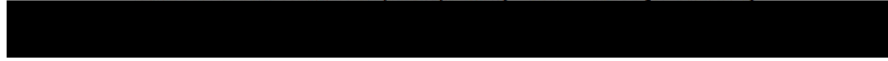
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	A large black rectangular area covering the majority of the table's content, indicating redacted information.	

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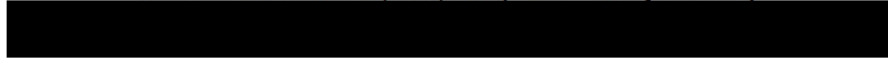
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	A large black rectangular area used for redaction, covering the entire content of the first row's right cell.
[1b2] wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features,	The Practicing '760 Products comprise the second image comprising a plurality of reflection features generated by the scene in response to illumination by the illumination features.

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	A large solid black rectangular redaction box covering the majority of the table's content area.
[1b3] wherein each of the reflection features comprises at least one beam profile,	The Practicing '760 Products comprise wherein each of the reflection features comprises at least one beam profile.

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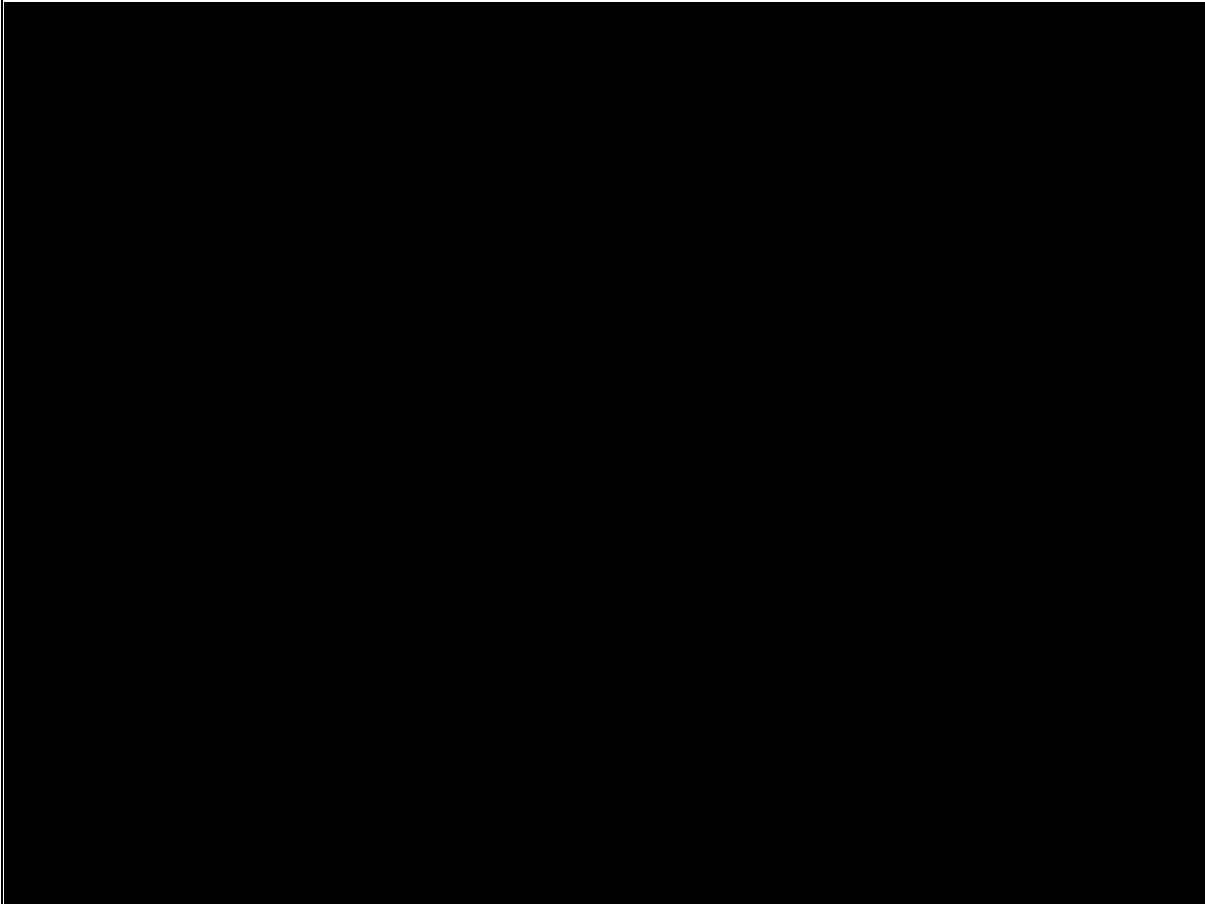
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[1b4] wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit,	The Practicing '760 Products comprise wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit.  A large solid black rectangular area used for redaction, covering the majority of the right-hand column of the table.
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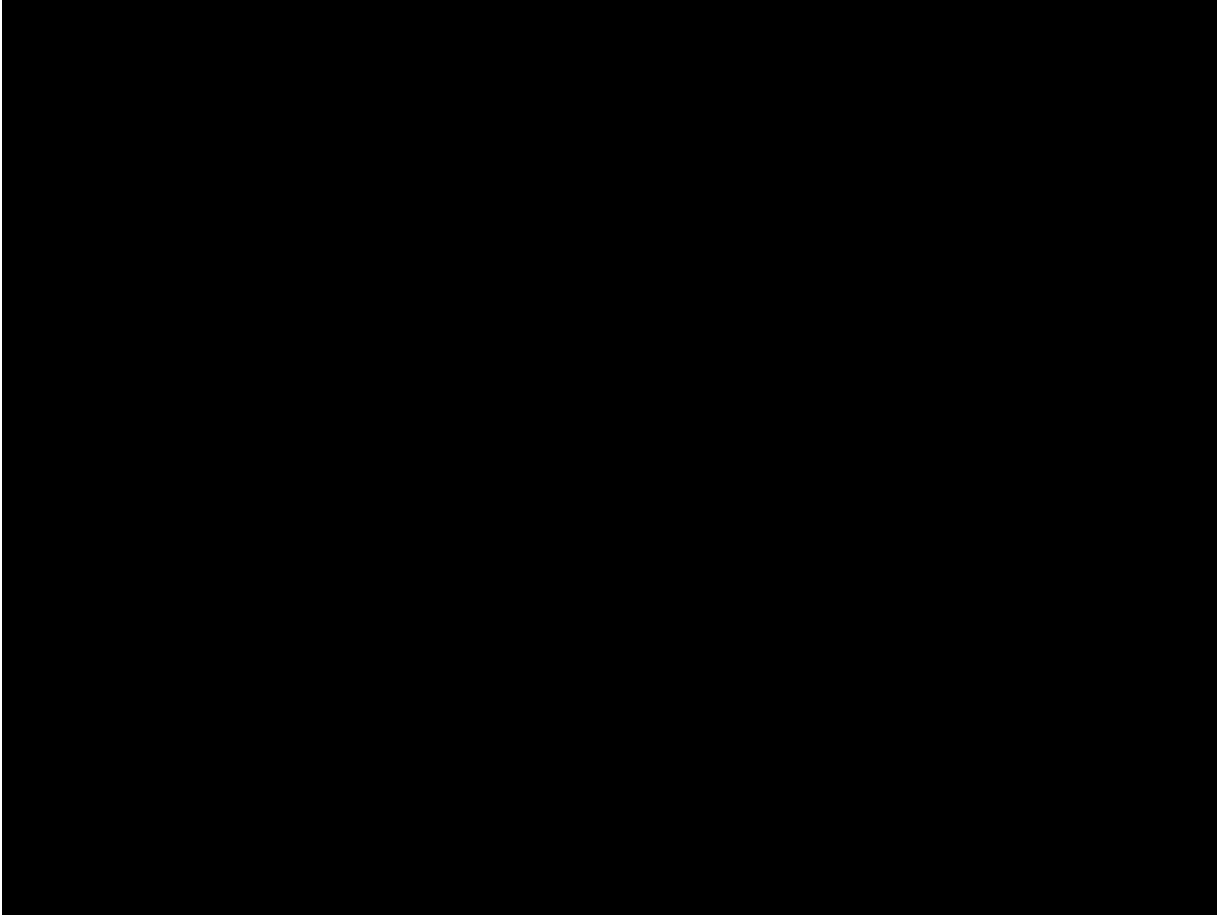
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[1b5] wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;	The Practicing '760 Products comprise wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin.  A large solid black rectangular area covering the majority of the right-hand cell, indicating redacted content.
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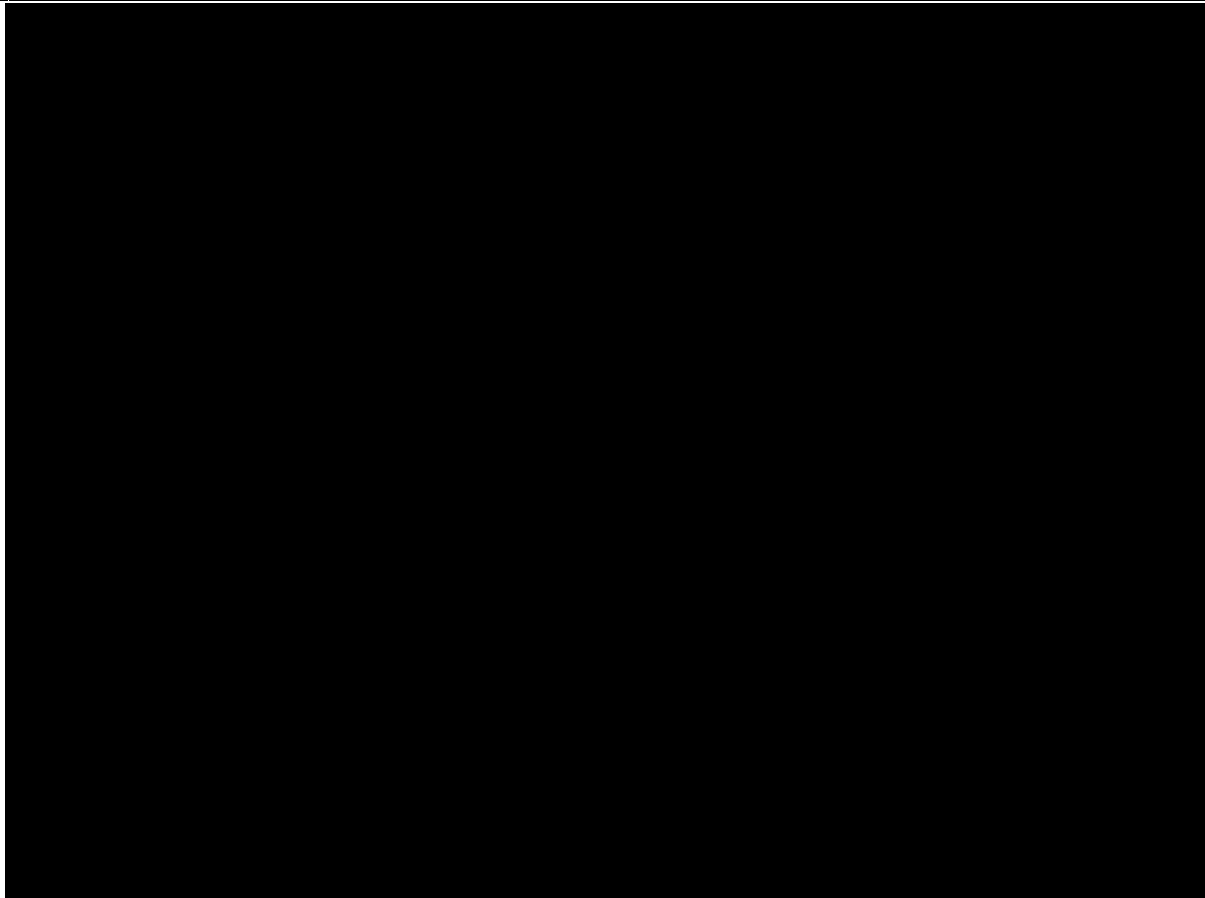


	A large solid black rectangular redaction box covering the entire content area of the table's first row.
[1c] at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the	The Practicing ’760 Products comprise at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects.

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processing unit, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects; and



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	A large solid black rectangular area used for redaction, covering the top half of the table's content area.
[1d] at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.	The Practicing '760 Products comprise at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.

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